
THE LOCK IS A DETERMINATION

Three Independent Axes, One Intersection, and What the Rows Were

SHORT COMMUNICATION

A geometric verdict kernel stated at theorem grade, its one constitutive blindness scoped, and its executable proof of load

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7 September 2026

ABSTRACT. A verification kernel reads three warrant rows over N contexts, forms their correlation Gram R , and reports whether the three axes enclose volume. We state what that report determines and what it does not, at the grade each claim carries. Three independent axes intersect in exactly one point: nondegeneracy is linear independence, so the determination is total about its object and no weaker reading survives. The kernel's scalar $\det R = \lambda^2$ is nonetheless blind to the sign of λ , and we show that this blindness is a fact about direction and never about the completeness of the determination, since the intersection is equally unique under every axis reflection. What varies between cases is not the strength of the lock but the provenance of its rows, and we give the three-way partition that follows: rows that are the closure itself, an axis left unpopulated so that no point is determined, and rows furnished by the world. A single-file Fortran implementation carries the theorems as procedures and its own battery of 248 computed checks as the proof of load; every figure quoted here was produced by that program at the stated build.

KEYWORDS Gram determinant, quaternion triple product, orientation blindness, rank tolerance, verification kernel, executable proof

1. STATEMENT

Let $q_1, q_2, q_3 \in \mathbb{R}^N$ be three rows, each centred, standardised, and normalised to unit length. Write Q for the $3 \times N$ matrix with these rows and

$$R = QQ^T \in \mathbb{R}^{3 \times 3}, \quad R_{ij} = \langle q_i, q_j \rangle,$$

for their correlation Gram. Let B be an orthonormal basis of the row span, fixed once, and let $c_i \in \mathbb{R}^3$ be the coordinates of q_i in B . Read each c_i as a pure quaternion $\hat{q}_i = 0 + c_{i1}\mathbf{i} + c_{i2}\mathbf{j} + c_{i3}\mathbf{k}$ and set

$$\lambda = \operatorname{Re}(\hat{q}_1 \hat{q}_2 \hat{q}_3).$$

Three facts are used throughout and none is new; the contribution is the arrangement and the typing. First, $\lambda = -\det[c_1; c_2; c_3]$, the scalar part of a product of three pure quaternions being the negative of the scalar triple product. Second, $R = CC^T$ with $C = [c_1; c_2; c_3]$, so $\det R = \lambda^2$. Third, $0 \leq \det R \leq 1$, Hadamard's inequality on unit rows.

We call the event $\det R > 0$ a **lock**. The paper is about what a lock determines.

2. THE DETERMINATION IS TOTAL

Read the three rows as normals of three planes through the origin of $\mathbb{R}^3$. Their intersection is the null space of C , of dimension $3 - \operatorname{rank} C$.

Theorem 1 (uniqueness). If $\det R > 0$ then $\operatorname{rank} C = 3$, the intersection has dimension zero, and the three planes meet in exactly one point. If $\operatorname{rank} C = 2$ the intersection is a line and no point is determined.

Proof. $\det R = (\det C)^2 > 0$ if and only if C is invertible, if and only if its null space is trivial. A null space of dimension one is a line through the origin. $\square$

The theorem is linear algebra and carries theorem grade. What it forbids is worth stating positively, because the vocabulary of verification tends to soften it. There is no second point. There is no interval of admissible points. Nothing about the determination is provisional. A lock is always a determination and is total about its object, and its object is its intersection.

The word *permission* has been used for the lock elsewhere in this programme, on the ground that a lock does not settle the proposition whose rows were read. That ground is correct and the word is wrong. It grades the lock, and locks carry no strengths. What differs between cases is what the rows were, which is Section 4.

3. THE ONE BLINDNESS, AND ITS SCOPE

Let $D = \operatorname{diag}(\pm 1, \pm 1, \pm 1)$ implement any pattern of axis reflections.

Theorem 2 (orientation blindness). $\det(DRD) = \det R$ for every D , while λ changes sign under every odd pattern.

Proof. $\det(DRD) = \det(D)^2 \det R = \det R$. Reflecting one row negates one c_i and hence the triple product. $\square$

The scalar the kernel reports cannot tell a proposition from its negation. This is real and it is the only blindness the kernel has. Its scope is exact: the blindness enters at the single operation $\lambda \mapsto \lambda^2$ and nowhere else. It is a fact about *direction*. It is not a fact about *completeness*: by Theorem 1 the intersection is equally unique under every reflection, so the determination is untouched by the operation that discards the sign. Direction is recovered from the rows themselves, never from the scalar, and in the implementation it is recovered from a linguistic seal that precedes the kernel and from the signed λ carried out of band.

At the executed build the reflection is exact and not merely small. Under strict IEEE arithmetic with reassociation disabled, negating one row negates every product and every sum bit for bit, and the power-of-two scaling used in the quaternion product commits no rounding, so the assertion $\lambda(\neg P) = -\lambda(P)$ is tested with exact equality and holds on three independent builds. A tolerance there would concede rounding the arithmetic does not commit.

4. WHAT THE ROWS WERE

Since the lock is total about its intersection, the only remaining question is what that intersection is an intersection of. Three cases exhaust it and none is a weaker lock.

Closure-rowed. The rows are the closure of the proposition's own terms: definitions fixed by stipulation and finite deductive objects delivered in hand. Then determining the rows and determining the proposition are one determination, not two, because there is no second thing for a gap to open between. The lock reaches the proposition. This is the case of a theorem of the algebra above: $\text{Re}(\mathbf{ijk}) = -1$ is closure-rowed.

Unpopulated. One axis carries no independent content. Then $\text{rank } C \leq 2$, the intersection is a line at best, and no lock forms. This is not a weak lock; it is the absence of one. An open conjecture over an unbounded domain with no proof in hand sits here.

World-rowed. The rows were furnished by measurement. The lock determines its intersection uniquely given those rows, and every revisability lives in the rows and none of it in the lock. A measurement that is later revised furnishes different rows; the lock on the old rows was never wrong about them.

The partition is structural rather than theorem-grade: it names the provenance of rows and not a property of the algebra. Its one theorem-grade leg is Theorem 1, which is what licenses saying that the

second case determines nothing at all.

5. READING RANK AT FINITE PRECISION

The implementation decides existence by dimension and not by determinant, because the dimension is the quantity Theorem 1 speaks of and the determinant is a magnitude read off it. Rank is read from the eigenvalues $w_1 \geq w_2 \geq w_3$ of R against

$$\tau = 8n\epsilon \max_i |w_i|,$$

with $n = 3$ and ϵ the unit roundoff. The tolerance is relative to the spectrum and carries no absolute floor. An earlier draft pinned a floor of one, and under it $R = 10^{-20}I$ returned rank zero where it is three; rank had become a fact about units. The relative form is invariant under uniform positive scaling of R , which is what a rank should be.

Two floors are read on R : the collapse floor $\epsilon_c = 100\epsilon N$ on $\det R$ and the conditioning gate on $\kappa(R) = w_1/w_3$. Their separation is known in closed form. Among correlation matrices of trace three at conditioning κ , the determinant is minimised by the positive-euicorrelation matrix and equals

$$\det R_{\min}(\kappa) = \frac{27\kappa}{(\kappa + 2)^3},$$

so the two floors cannot contend while $\kappa < \sqrt{27/(100\epsilon N)}$, which at $N = 24$ is 7.1×10^6 , above the gate at 10^6 . Within that envelope the verdict boundary is decided by structure and not by rounding. Where the rank floor and the collapse floor nonetheless disagree, the implementation neither declares the lock broken nor declares it determined; it reports the arithmetic as incomplete and re-runs at higher precision. A floor disagreement escalates and never decides.

6. FALSIFICATION

Three criteria, each aimed at one theorem of this paper and each cheap to run against the shipped program.

1. Exhibit three independent unit rows whose planes meet in more than one point, or in an interval. This falsifies Theorem 1 and with it the spine.
2. Under a strict IEEE build with reassociation disabled, exhibit one row reflection under which the composed scalar fails $\lambda(\neg P) = -\lambda(P)$ bit for bit. This falsifies the exactness claim of Section 3.
3. Exhibit a positive semi-definite Gram whose computed rank changes under uniform positive scaling within the representable range. This falsifies the scale invariance of Section 5.

7. EXECUTED RECEIPTS

Every figure below was printed by the shipped program, built as `-std=f2018 -O2 -fno-fast-math`

-ffp-contract=off, and is local execution evidence gated against its bound; the identity residual differs by half an ulp between builds and both sides pass the printed bound.

On a three-axis warrant matrix over $N = 24$ contexts, intersection dimension 0, $\det R = 0.844905725419$, $|\lambda^2 - \det R| = 1.110 \times 10^{-16}$ against the conditioning-scaled bound 1.996×10^{-15} , rank margin 8.350×10^{13} . With the third axis made a combination of the first two, intersection dimension 1, verdict broken, and the message names the geometry: rank two, the intersection is a line and not a point.

Under full negation, $\lambda(P) = -0.919187535500$ and $\lambda(\neg P) = +0.919187535500$, with $\max |G(P) - G(\neg P)| = 0.000 \times 10^0$ and $|\det R(P) - \det R(\neg P)| = 0.000 \times 10^0$ exactly. The Hamilton relation returns $\text{Re}(\mathbf{ijk}) = -1.000000000000$. On the orthonormal Fourier triad $\{\sin t, \cos t, \sin 2t\}$ the lock reaches its ceiling, $\det R = 1.000000000000$ and $|\lambda| = 1.000000000000$.

The battery of 248 computed checks, eight classes over every public procedure including hostile non-finite and extreme-magnitude input, returned zero failures on three builds: strict IEEE with floating-point traps armed and signalling-NaN initialisation, optimised, and optimised with contraction disabled. The program stops with a nonzero exit if any check fails, so a binary that prints its final line has passed.

8. CONCLUSION

A lock determines its intersection completely and uniquely, and the one thing it cannot read is the sign. Those are two different facts, and conflating them produced a vocabulary that graded locks by strength. Locks carry no strengths. The compartment names what the rows were, and only in the closure-rowed case does the determination reach the proposition, because only there are the rows and the proposition the same thing.

9. PROVENANCE AND METHOD DISCLOSURE

This paper was developed under Trisduction, a verification and organizing discipline that adds the cited results no warrant; the method and its executable batteries are stated in full at the reference below.

10. REFERENCES

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